CS 133 — DATA VISUALIZATION PROJECT PROPOSAL

Will AI Replace Us?

Mapping how data scientists talk about AI-driven job displacement on LinkedIn

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THE PROBLEM

AI job-loss anxiety is rising — and the experts building AI are talking about it too

Layoffs are increasingly blamed on AI, and job insecurity is a known source of stress. Data scientists and AI practitioners are uniquely placed to weigh in — they build and understand these tools better than most. As students about to enter this same job market, the question is personal for us too.

Source: LinkedIn discourse data, April 2024 – June 2025

The Dataset

784
LinkedIn posts, Apr 2024 – Jun 2025

7
LDA topics, already scored per post

CC0
Open access, CORA / Dataverse CSUC

The 7 topics include

Will AI replace us? · LLM agents at work · AutoML automating ML · Vibe coding with GenAI · Juniors & citizen devs

Samokhina, O. (2026). Hipertext.net, 32 — full citation on References slide

Exploratory Questions

Plot type	Question
Bar chart	Which topics have the highest share of anxious posts?
Scatter	Do anxious posts also lean optimistic, or does each post pick a side?
Count plot	What is the overall tone split, and does it shift by topic?
Interactive line (Plotly)	Does emotional language fade as posts get longer?
Small multiples	Do longer posts blend more topics instead of staying on one?

5 questions, 5 plot types, all run on the real data

Strategy

Tools

Python in Colab · pandas · seaborn / plotly · scikit-learn

Algorithms

LDA (provided) · lexicon tone label already built (Dan) · deciding whether to upgrade to transformer / zero-shot · logistic regression for prediction

Prediction target

Can topic profile plus post length predict sentiment category?

Goal: find out whether AI and data-science practitioners sound more anxious or more confident — and whether that depends on which specific AI capability they’re discussing.

Data source: 784 LinkedIn posts, CORA / Dataverse CSUC (CC0), pre-scored on 7 LDA topics — see References.

Expected Results & Testing

WE EXPECT

- Topics skew toward “Will AI replace us?” and “LLM agents at work”
- Mixed, ambivalent sentiment — not uniformly negative
- We’ll test whether longer posts mix more topics
- A modest, real lift over a majority-class baseline — not a strong classifier

HOW WE WILL TEST IT

- 80/20 split (~627 train / ~157 test), plus k-fold CV on the training set to compare models before touching test
- Accuracy and F1 against the majority baseline
- Re-check findings after removing outlier-length posts
- Manual spot-check of a sample of labels

References

Samokhina, O. (2026). *Technological optimism or automation anxiety: how data scientists discuss AI-driven job displacement.* Hipertext.net, 32.

Dataset: CORA / Dataverse CSUC, DOI 10.34810/DATA3307. CC0 license. Based on a Master's dissertation (DiCEM, Universitat Pompeu Fabra, Barcelona, 2024–2025).

Blei, D., Ng, A., & Jordan, M. (2003). Latent Dirichlet Allocation. *Journal of Machine Learning Research.*

[Team: add any other papers or repos each of you read]